Lecture 11

Singularities and residues

Objectives. *An isolated singularity contributes one Laurent coefficient to a contour integral. We prove this residue formula and use it to evaluate real integrals. The limiting contours require estimates as well as residues.*

By the end of this lecture you will be able to:

- Expand a holomorphic function on an annulus and classify an isolated singularity.
- Evaluate residues and justify semicircle and keyhole limits.
- Count zeros minus poles with multiplicities by the argument principle.
- Characterise meromorphic functions on the Riemann sphere.

11.1 Laurent series and isolated singularities

An annulus is $A(a;r,R)=\{z : r<|z-a|<R\}$, where $0\le r<R\le\infty$. It excludes its centre even when $r=0$.

Theorem 11.1 (Laurent expansion). *If f is holomorphic on $A(a;r,R)$, then it has the expansion*

$$f(z)=\sum_{n\in\mathbb{Z}} a_n(z-a)^n,$$
$$a_n=\frac{1}{2\pi\mathrm{i}}\oint_{|\zeta-a|=s}\frac{f(\zeta)}{(\zeta-a)^{n+1}}\,\mathrm{d}\zeta. \tag{11.1}$$

The coefficient radius satisfies $r<s<R$. The positive and negative parts converge absolutely and uniformly on each compact subannulus. The expansion is unique among series whose positive and negative parts converge throughout the annulus. All coefficient circles are counter-clockwise.

Proof. Translate so that $a=0$. Choose $r<r_1<|z|<r_2<R$. Remove a small closed disk about z from the closed annulus $r_1\le|\zeta|\le r_2$. Boundary cancellation (Lemma 10.1) applied to $f(\zeta)/(\zeta-z)$, followed by the small-circle limit from Cauchy's formula, gives

$$f(z)=\frac{1}{2\pi\mathrm{i}}\oint_{|\zeta|=r_2}\frac{f(\zeta)}{\zeta-z}\,\mathrm{d}\zeta-\frac{1}{2\pi\mathrm{i}}\oint_{|\zeta|=r_1}\frac{f(\zeta)}{\zeta-z}\,\mathrm{d}\zeta.$$

On the outer circle expand in z/ζ. On the inner circle expand in ζ/z, with the minus sign already present:

$$\frac{1}{\zeta-z}=\sum_{n=0}^{\infty}\frac{z^n}{\zeta^{n+1}},\qquad -\frac{1}{\zeta-z}=\sum_{m=0}^{\infty}\frac{\zeta^m}{z^{m+1}}.$$

On any closed subannulus strictly between r_1 and r_2, both ratios are bounded by numbers smaller than one. The geometric majorants prove absolute uniform convergence and allow termwise integration. These are the required positive and negative series.

For each integer n, the function $f(\zeta)/\zeta^{n+1}$ is holomorphic between any two admissible coefficient circles. Boundary cancellation therefore makes its circle integral independent of the radius. Every compact subannulus can be enclosed by r_1 and r_2, so the expansion holds throughout the original annulus. For any competing series whose two parts converge there, apply the uniform power-series convergence of Theorem 1.5 to the positive part in z and the negative part in $1/z$. Both converge uniformly on each interior circle. Integrate that series term by term after multiplying by z^{-n-1}. The integral of z^j is zero unless $j = -1$, when it is $2\pi\mathrm{i}$. This recovers a_n and proves uniqueness. ∎

The annulus must contain no pole of f. Poles may lie in its excluded hole. For example,

$$\frac{1}{z(z-2)} = -\frac{1}{2z} + \frac{1}{2(z-2)} = -\frac{1}{2z} - \sum_{n=0}^{\infty} \frac{z^n}{2^{n+2}}, \qquad 0 < |z| < 2.$$

The pole at 2 bounds the outer radius. A different Laurent expansion holds for $|z| > 2$.

An *isolated singularity* at a means that f is holomorphic on some punctured disk $0 < |z-a| < R$. It is *removable* if f extends holomorphically to a. It is a *pole of order* m if $f(z) = (z-a)^{-m}g(z)$ near a, where $m \geq 1$ and g is holomorphic with $g(a) \neq 0$. Every other isolated singularity is *essential*.

By Laurent uniqueness, the removable case has no negative coefficients. A pole of order m has $a_{-m} \neq 0$ and no terms below $-m$. Conversely, either of these coefficient conditions gives the stated extension or factorisation. Thus an essential singularity has infinitely many nonzero negative coefficients. The function $e^{1/z}$ is an example.

Theorem 11.2 (Removable singularity). *A function holomorphic and bounded on a punctured disk has a removable singularity at its centre.*

Proof. If $|f| \leq M$ there, the coefficient formula and the ML-estimate, Equation (9.1), give $|a_{-m}| \leq Ms^m$ for every $m \geq 1$ and every sufficiently small $s > 0$. Letting $s \downarrow 0$ kills every negative coefficient. The remaining power series supplies the extension. ∎

11.2 The residue theorem

The *residue* at an isolated singularity a is $\operatorname{Res}_{z=a} f = a_{-1}$. A function is *meromorphic* on an open set if it is holomorphic except for isolated poles with no accumulation point in that open set.

Theorem 11.3 (Residue theorem). *Let E have the finite line-and-circle boundary of Lemma 10.1, oriented with E on the left. Suppose f is holomorphic on a neighbourhood of $\overline{E}$ except for finitely many isolated singularities $a_1, \dots, a_N$ in E. Then*

$$\int_{\partial E} f(z)\,\mathrm{d}z = 2\pi\mathrm{i} \sum_{j=1}^{N} \operatorname{Res}_{z=a_j} f.$$

Proof. Remove disjoint small closed disks about the singularities, with all disks inside E. The remaining compact region lies in the open set of holomorphy. Boundary cancellation equates its original boundary integral to the sum of counter-clockwise integrals on these small circles. Uniform Laurent convergence allows integration term by term on each circle. Only the power -1 contributes, giving $2\pi\mathrm{i}a_{-1}$. ∎

For a simple pole of u/v, where u and v are holomorphic, $v(a) = 0$ and $v'(a) \neq 0$, factor $v(z) = (z-a)\widetilde{v}(z)$ with $\widetilde{v}(a) = v'(a)$. This gives

$$\operatorname{Res}_{z=a} \frac{u(z)}{v(z)} = \frac{u(a)}{v'(a)}.$$

For a pole of order m, Taylor expansion of the holomorphic extension of $(z-a)^m f(z)$ gives

$$\operatorname{Res}_{z=a} f = \frac{1}{(m-1)!} \frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}} \big((z-a)^m f(z)\big)\Big|_{z=a}.$$

For example, e^z/z^2 has residue 1 at 0, since the first derivative of e^z there is 1.

11.3 Rational and trigonometric integrals

For $k \geq 0$, integrate $e^{\mathrm{i}kz}/(1+z^2)$ along the real segment from $-R$ to R and the upper semicircle from R to $-R$, where $R > 1$. The only enclosed pole is i, with residue $e^{-k}/(2\mathrm{i})$. On the semicircle $\left|e^{\mathrm{i}kz}\right| \leq 1$ and $\left|1+z^2\right| \geq R^2 - 1$. Thus, by the ML-estimate, Equation (9.1), its integral has modulus at most $\pi R/(R^2-1)$ and tends to zero. Absolute convergence on the real line follows from $\left|e^{\mathrm{i}kx}\right| = 1$ and the integrable denominator. For $k < 0$, use the lower semicircle from R to $-R$. This closes clockwise and encloses $-\mathrm{i}$, with residue $e^k/(-2\mathrm{i})$. The clockwise residue factor is $-2\pi\mathrm{i}$. Hence, for every $k \in \mathbb{R}$,

$$\int_{-\infty}^{\infty} \frac{e^{\mathrm{i}kx}}{1+x^2}\,\mathrm{d}x = \pi e^{-|k|}. \tag{11.2}$$

The case $k = 0$ is the rational integral $\int_{\mathbb{R}} (1+x^2)^{-1}\,\mathrm{d}x = \pi$.

These finite contours lie in an open set containing the real boundary segments and semicircles after the poles are removed. Small disks about the poles lie wholly inside the half-disks. The residue proof therefore uses boundary cancellation on a compact region inside its open holomorphic set. It does not apply the open half-disk theorem directly to its boundary.

In INT's convention the displayed integral is its transform. In our unitary spatial convention, whose kernel is $e^{-\mathrm{i}kx}$,

$$\mathcal{F}\left(\frac{1}{1+x^2}\right)(k) = \sqrt{\frac{\pi}{2}} e^{-|k|}.$$

The sign reversal has no effect here because the answer is even in k.

For a trigonometric example let $b > 1$ and set $z = e^{\mathrm{i}t}$. The counter-clockwise unit circle gives

$$\int_0^{2\pi} \frac{\mathrm{d}t}{b+\cos t} = \oint_{|z|=1} \frac{2\,\mathrm{d}z}{\mathrm{i}(z^2+2bz+1)} = \frac{2\pi}{\sqrt{b^2-1}}.$$

Indeed, the roots are $-b+\sqrt{b^2-1}$ and $-b-\sqrt{b^2-1}$. Only the first lies inside the unit circle: its modulus is $1/(b+\sqrt{b^2-1}) < 1$. The residue there is $1/(\mathrm{i}\sqrt{b^2-1})$, which proves the last equality.

11.4 Jordan's lemma and a conditional integral

The ML bound alone does not suffice when the rational factor decays only as $1/z$. The exponential supplies a stronger estimate.

Lemma 11.4 (Jordan). *Let $k > 0$ and let g be continuous on the upper semicircle C_R. Writing $M_R = \max_{C_R} |g|$, we have*

$$\left| \int_{C_R} e^{\mathrm{i}kz} g(z)\, \mathrm{d}z \right| \leq \frac{\pi M_R}{k}.$$

In particular the integral tends to zero if $M_R \to 0$. For $k < 0$ the corresponding statement holds on the lower semicircle, with $|k|$ in the denominator.

Proof. Parametrise $z = Re^{\mathrm{i}t}$ with $0 \leq t \leq \pi$. The modulus of the exponential is $e^{-kR\sin t}$. On $[0, \pi/2]$, concavity gives $\sin t \geq 2t/\pi$. Reflection about $\pi/2$ therefore gives

$$RM_R \int_0^{\pi} e^{-kR\sin t}\, \mathrm{d}t \leq 2RM_R \int_0^{\pi/2} e^{-2kRt/\pi}\, \mathrm{d}t \leq \frac{\pi M_R}{k}.$$

Reflecting the semicircle across the real axis proves the lower version. ∎

Let $a > 0$ and $k > 0$. Apply the lemma to $g(z) = z/(z^2 + a^2)$. For $R > a$, its bound is $M_R \leq R/(R^2 - a^2)$, which tends to zero. The residue of $ze^{\mathrm{i}kz}/(z^2 + a^2)$ at $\mathrm{i}a$ is $e^{-ak}/2$. The upper contour therefore gives $\mathrm{i}\pi e^{-ak}$. For $k < 0$ the lower contour is clockwise. Its pole at $-\mathrm{i}a$ has residue $e^{-a|k|}/2$, giving the negative of that answer:

$$\int_{-\infty}^{\infty} \frac{xe^{\mathrm{i}kx}}{x^2 + a^2}\, \mathrm{d}x = \mathrm{i}\pi \operatorname{sgn}(k) e^{-a|k|}, \qquad k \neq 0. \tag{11.3}$$

The integral here is improper with each one-sided tail convergent. To check this rather than merely a symmetric limit, put $q(x) = x/(x^2 + a^2)$. On either tail, integration by parts against $e^{\mathrm{i}kx}$ bounds an integral from A to B by

$$\frac{1}{|k|} \left(|q(A)| + |q(B)| + \int_A^B |q'(x)|\, \mathrm{d}x \right).$$

The derivative is $q'(x) = (a^2 - x^2)/(x^2 + a^2)^2$. Thus $q(x) = O(1/|x|)$ and $q'(x) = O(1/x^2)$, so the bound tends to zero when both endpoints go to the same infinity. This proves each tail's Cauchy criterion. The absolute integral diverges, since $\left|xe^{\mathrm{i}kx}/(x^2 + a^2)\right| \sim 1/|x|$.

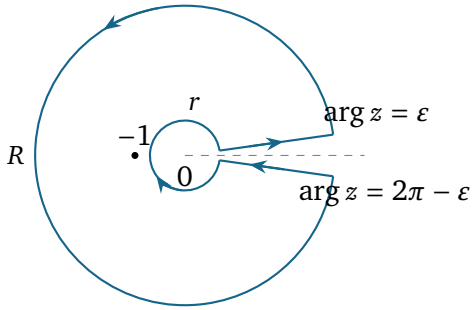
11.5 A branch power and the keyhole

For this section use the *keyhole branch* with cut $[0, \infty)$:

$$\log_+(z) = \ln|z| + \mathrm{i}\arg z, \qquad 0 < \arg z < 2\pi, \qquad z^{a-1} = e^{(a-1)\log_+(z)}, \qquad 0 < a < 1.$$

This is a named change from the principal logarithm of § 9.2. The same local polar differentiation gives $(\log_+)'(z) = 1/z$. Consider $f(z) = z^{a-1}/(1 + z)$. Its only pole in the slit plane is -1, with residue $e^{\mathrm{i}\pi(a-1)}$.

Use an annular sector $r < |z| < R$ with $\varepsilon < \arg z < 2\pi - \varepsilon$, where $r < 1 < R$ and $0 < \varepsilon < \pi/2$. Its boundary follows the upper ray outwards, the outer circle counter-clockwise, the lower ray inwards and the inner circle clockwise. All its points lie off the cut. Remove a small disk about -1 inside the sector and apply the residue theorem.



The dashed ray is excluded. The two straight sides remain inside the slit plane until their limiting values are taken.

For fixed r and R, let $\varepsilon \downarrow 0$. The ray integrands converge uniformly on $[r, R]$ to their one-sided boundary values. The upper ray contributes $I_{r,R} = \int_r^R x^{a-1}/(1+x)\,\mathrm{d}x$. The inward lower ray contributes $-e^{2\pi \mathrm{i} a} I_{r,R}$. The extra factor $e^{2\pi \mathrm{i}}$ from $\mathrm{d}z$ is 1. The circular pieces converge to the corresponding full-circle integrals with these limiting branch values. For $R > 2$ their outer contribution has modulus at most $2\pi R^a/(R-1)$; for $r < 1/2$ the inner bound is $2\pi r^a/(1-r)$. Both tend to zero at the required endpoints. The real integral converges because its integrand is $O(x^{a-1})$ at zero and $O(x^{a-2})$ at infinity. Hence the residue identity becomes

$$(1 - e^{2\pi \mathrm{i} a}) \int_0^\infty \frac{x^{a-1}}{1+x}\,\mathrm{d}x = 2\pi \mathrm{i} e^{\mathrm{i}\pi(a-1)}.$$

Using $1 - e^{2\pi \mathrm{i} a} = -2\mathrm{i} e^{\pi \mathrm{i} a} \sin(\pi a)$ gives

$$\int_0^\infty \frac{x^{a-1}}{1+x}\,\mathrm{d}x = \frac{\pi}{\sin \pi a}, \qquad 0 < a < 1. \tag{11.4}$$

Lecture 13 uses this integral to prove Gamma reflection.

11.6 Counting zeros and the Riemann sphere

Theorem 11.5 (Argument principle). *Let f be meromorphic on a neighbourhood of the closure of a region E of the residue theorem. Suppose it has no zero or pole on ∂E and is not identically zero on any component meeting $\overline{E}$. Then*

$$\frac{1}{2\pi \mathrm{i}} \int_{\partial E} \frac{f'(z)}{f(z)}\,\mathrm{d}z = N - P,$$

where N counts zeros in E and P counts poles in E, both with multiplicity and with the positive boundary orientation.

Proof. Zeros have the local factorisation from § 10.4. The identity theorem, Theorem 10.6, makes each zero isolated unless the function vanishes on its component. Poles are isolated by meromorphy. There are finitely many of each in $\overline{E}$: otherwise compactness would give an accumulation point there, contrary to the local factorisations and the absence of boundary zeros and poles. Near a zero of order m or a pole of order $-m$, write $f(z) = (z-a)^m g(z)$ with g nonzero and holomorphic. Here m is a nonzero integer, negative for a pole. Differentiation gives

$$\frac{f'(z)}{f(z)} = \frac{m}{z-a} + \frac{g'(z)}{g(z)}.$$

The second term is holomorphic. The residue is therefore m. The zeros and poles are discrete in the union of the original neighbourhood components meeting $\overline{E}$, and none lies on the compact boundary. Shrink the neighbourhood of $\overline{E}$ to exclude any zeros and poles outside it. The residue theorem for f'/f then gives precisely $N - P$. ∎

For example, on $|z| = 2$ the function $f(z) = (z-1)^3/z^2$ has three zeros and two poles counted with multiplicity. Its logarithmic derivative is $3/(z-1) - 2/z$. The integral divided by $2\pi\mathrm{i}$ is therefore 1, although the total number of distinct zeros and poles is two. The winding number of a closed curve γ about a point a off it is defined by $n(\gamma, a) = (2\pi\mathrm{i})^{-1} \int_\gamma (z-a)^{-1}\,\mathrm{d}z$. The argument principle computes this integral for the image curve $f \circ \gamma$ when $\gamma = \partial E$ is a single boundary component.

The *Riemann sphere* is $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$, described by the coordinate z away from infinity and $w = 1/z$ near infinity. A neighbourhood of infinity contains $\{z : |z| > R\} \cup \{\infty\}$ for some $R > 0$. A function meromorphic on the sphere means a function meromorphic in the finite plane for which $f(1/w)$ has a removable singularity or a pole at $w = 0$. Constant finite functions are included.

Theorem 11.6. *The meromorphic functions on the Riemann sphere are exactly the rational functions.*

Proof. First let $f = p/q$ be rational. Cancel common factors. The finite singularities are poles at zeros of q. If the degrees are j and m, then $f(1/w)$ is w^{m-j} times a quotient of polynomials in w whose values at zero are the nonzero leading coefficients. Thus infinity is removable if $j \le m$ and a pole if $j > m$. The zero rational function is removable everywhere.

Conversely, meromorphy at infinity supplies an exterior region with no finite poles. In the remaining compact disk there can be only finitely many poles, since each point has a neighbourhood with at most one pole. Subtract their finite Laurent principal parts from f. The result H extends to an entire function. At infinity, write the finite principal part of $f(1/w)$ as $\sum_{j=1}^{m} c_j w^{-j}$, with an empty sum in the removable case. Subtract the polynomial $Q(z) = \sum_{j=1}^{m} c_j z^j$ as well. The function $H - Q$ is entire and bounded for large $|z|$: the remainder at infinity is bounded, and each subtracted finite-pole principal part tends to zero there. It is bounded on the remaining disk by continuity. Liouville's theorem, Theorem 10.4, says that a bounded entire function is constant. It therefore makes $H - Q$ constant. Reassembling f expresses it as a polynomial plus finitely many principal parts, hence as a rational function. ∎

Summary.

- A holomorphic function on an annulus has a unique Laurent expansion. Its coefficient of power -1 is the residue.
- Positive boundary integrals equal $2\pi\mathrm{i}$ times the enclosed residues for the finite line-and-circle contours proved here.
- A limiting contour requires a vanishing arc estimate. Jordan's lemma treats exponential kernels with a decaying amplitude.
- The keyhole branch with argument in $(0, 2\pi)$ gives $\int_0^\infty x^{a-1}/(1+x)\,\mathrm{d}x = \pi/\sin\pi a$ for $0 < a < 1$.
- The argument principle counts zeros minus poles with multiplicity. Meromorphic functions on the sphere are rational.

Lecture 12 applies the same oriented boundary integrals to analytic continuation across an integration interval.

References

J. Bak and D. J. Newman, *Complex Analysis*, third edition, Springer, 2010, §9.2, pp. 120–125. E. M. Stein and R. Shakarchi, *Complex Analysis*, Princeton, 2003, Chapter 3, §§1–4, pp. 73–90. These sources' unnormalised contour integrals are used as written. The Fourier conversion above uses the frozen unitary convention.